

Variables:

Variable is a container is used to store the values in our programs.

Variable is a named memory [bytes] location where we can store and manipulate [modify] the values in our programs.

Each and every variable is stored in primary memory i.e. RAM. Due to this the variables are deleted automatically once the program / function execution is completed.

Variables are case sensitive. i.e. lower and upper are different.

Eg:

```
Int a=10;
```

```
Int A=20;
```

Every variable having 2 stages.

1. Variable declaration / declared

Eg: `int a;`

2. Variable initialization / defined

`a=100;`

in C compiler we should have to declare the variable in the first line of any program / function only. In C++ we can declare anywhere.

Syntax:

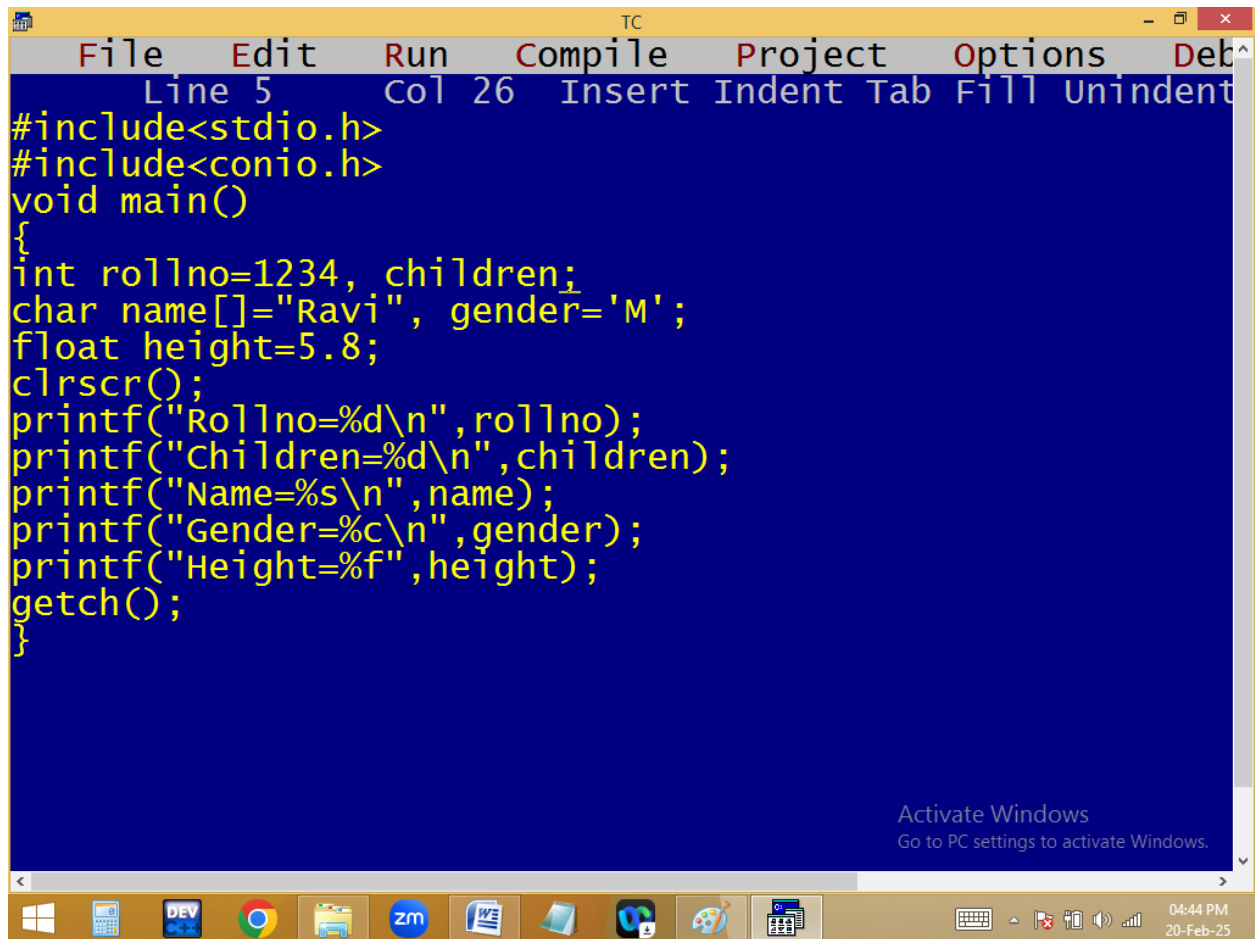
Datatype variable[=**value**], variable[=**value**],...;

Eg:

```
int rollno=1234;
```

```
char name[ ]="Ravi", gender='M';
```

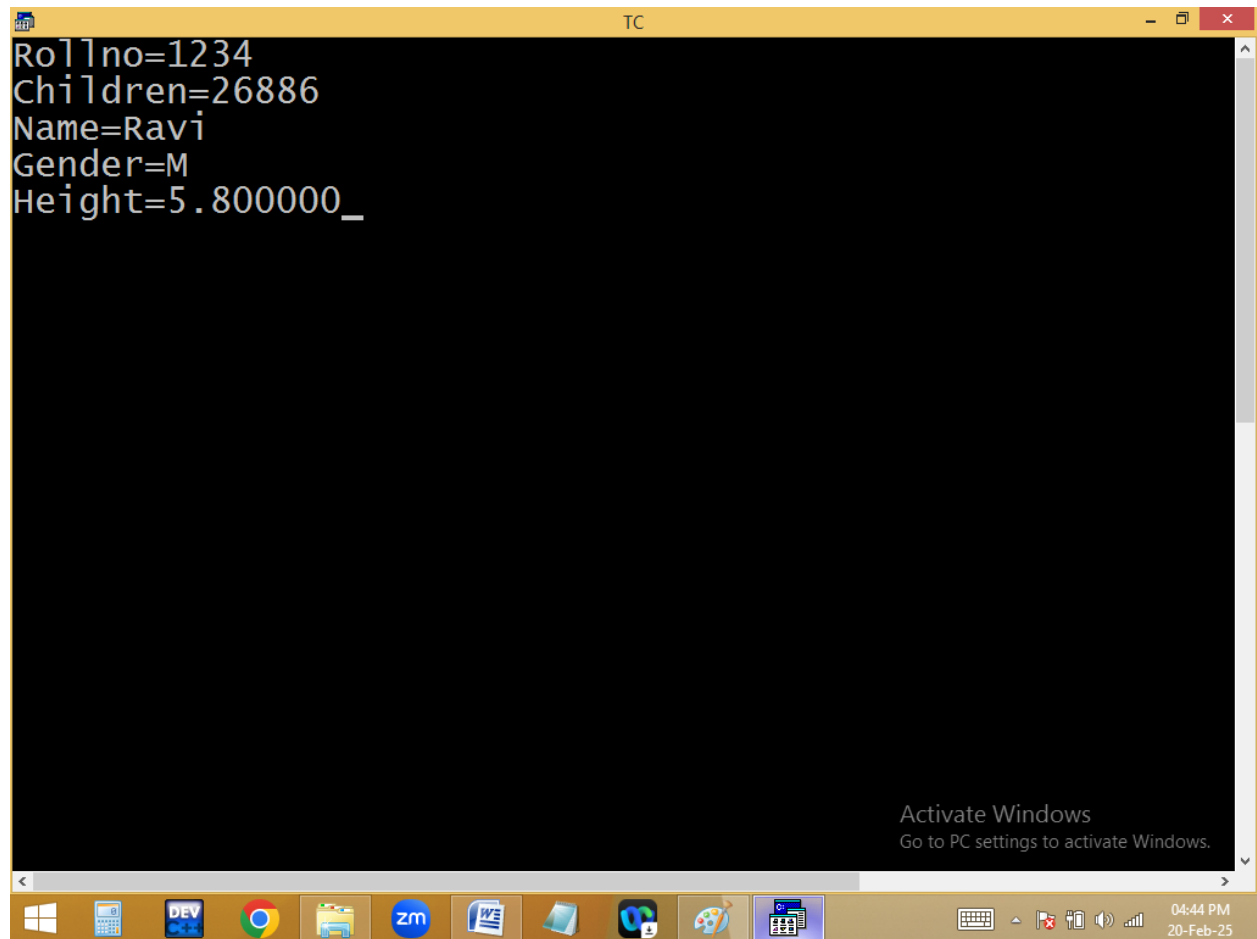
```
Float height = 5.8;
```

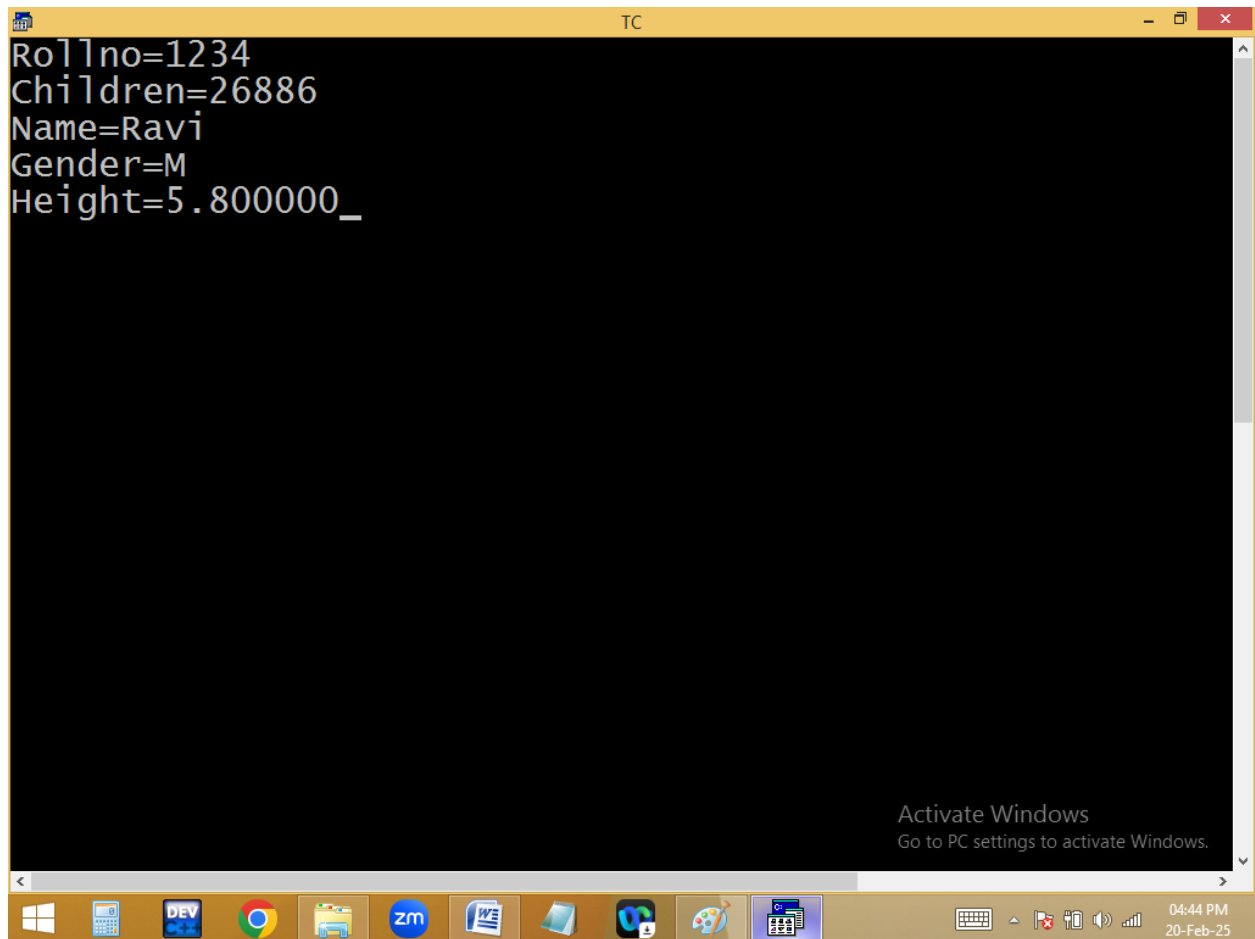


```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 26 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno=1234, children;
char name[]="Ravi", gender='M';
float height=5.8;
clrscr();
printf("Rollno=%d\n",rollno);
printf("Children=%d\n",children);
printf("Name=%s\n",name);
printf("Gender=%c\n",gender);
printf("Height=%f",height);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

04:44 PM
20-Feb-25





A screenshot of a Turbo C++ (TC) window. The window title bar is yellow and contains the text 'TC'. The main area is black with white text. The text displayed is:

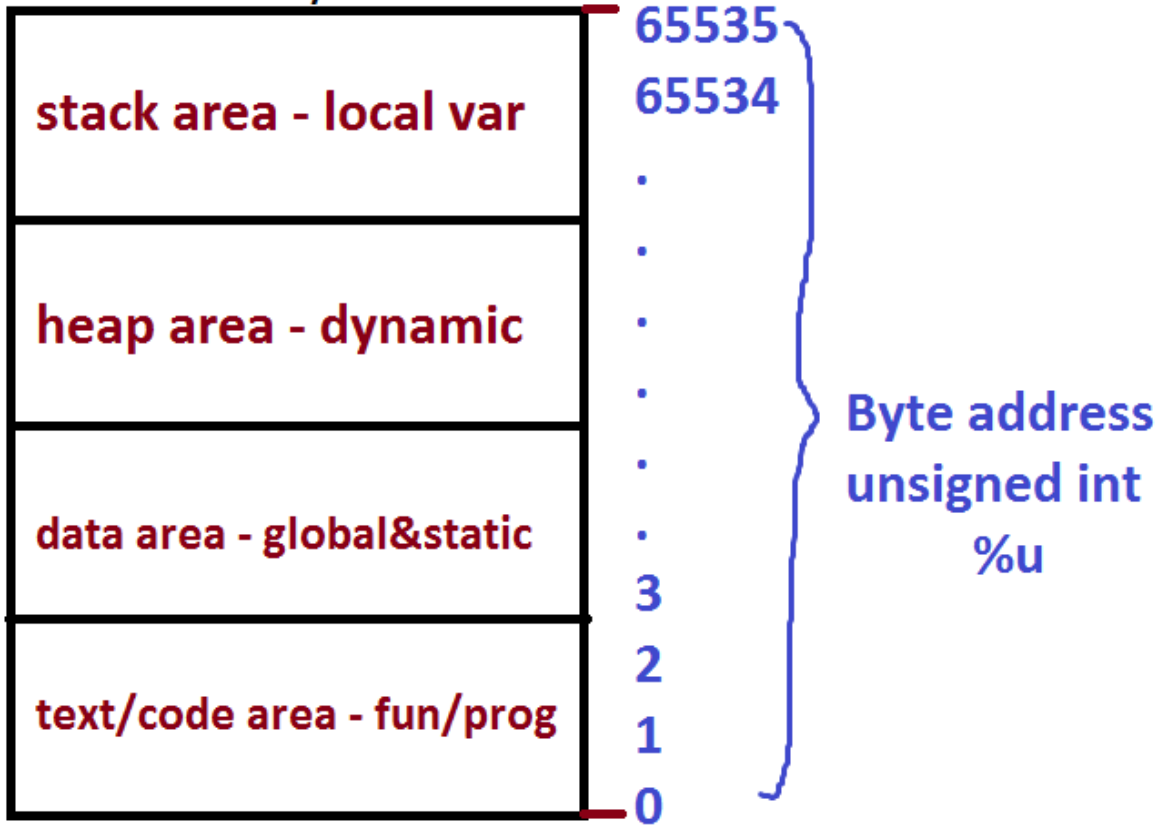
```
Rollno=1234  
Children=26886  
Name=Ravi  
Gender=M  
Height=5.800000_
```

In the bottom right corner of the black area, there is a watermark that says 'Activate Windows Go to PC settings to activate Windows.' The taskbar at the bottom is light blue and contains several icons: Windows logo, calculator, DEV, Chrome, File Explorer, Zoom, Word, a folder icon, a game icon, and a calculator icon. On the right side of the taskbar, there are system icons for keyboard, volume, and network, along with the date and time: '04:44 PM 20-Feb-25'.

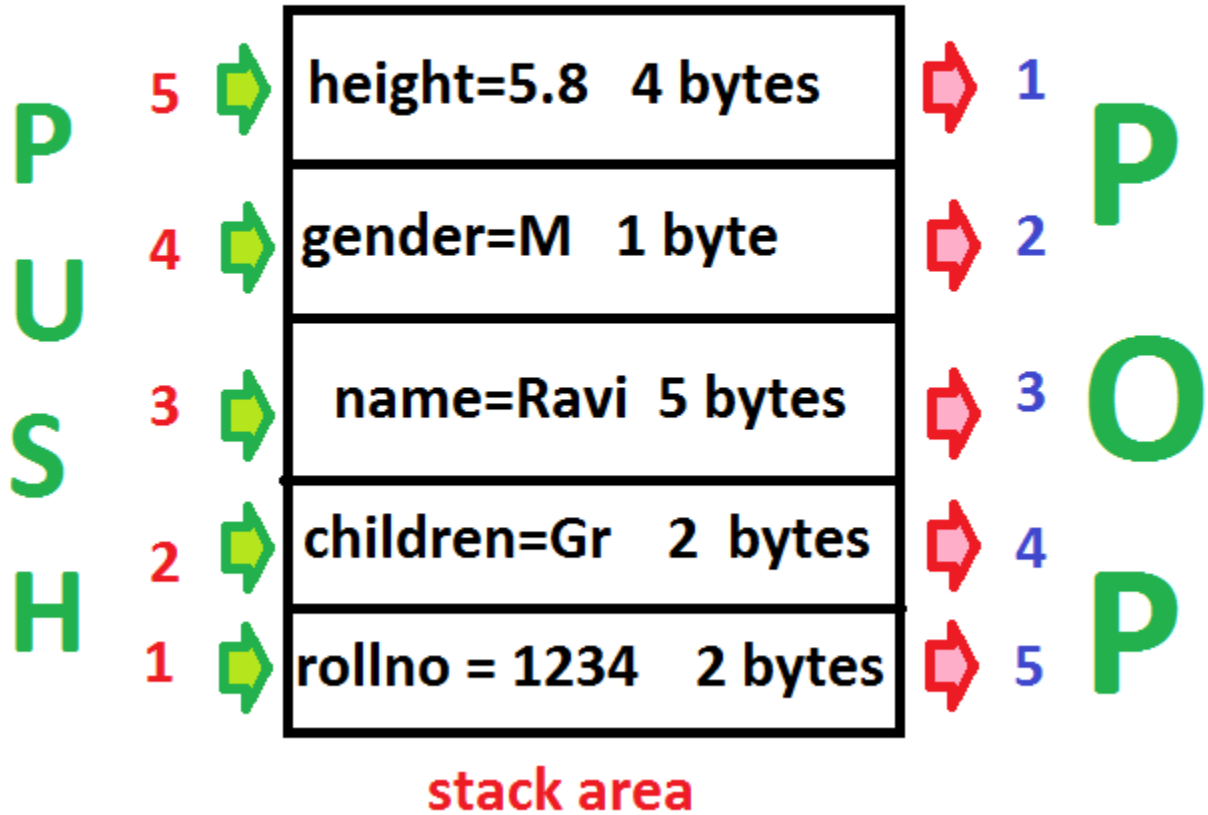
Memory allocation for variables:

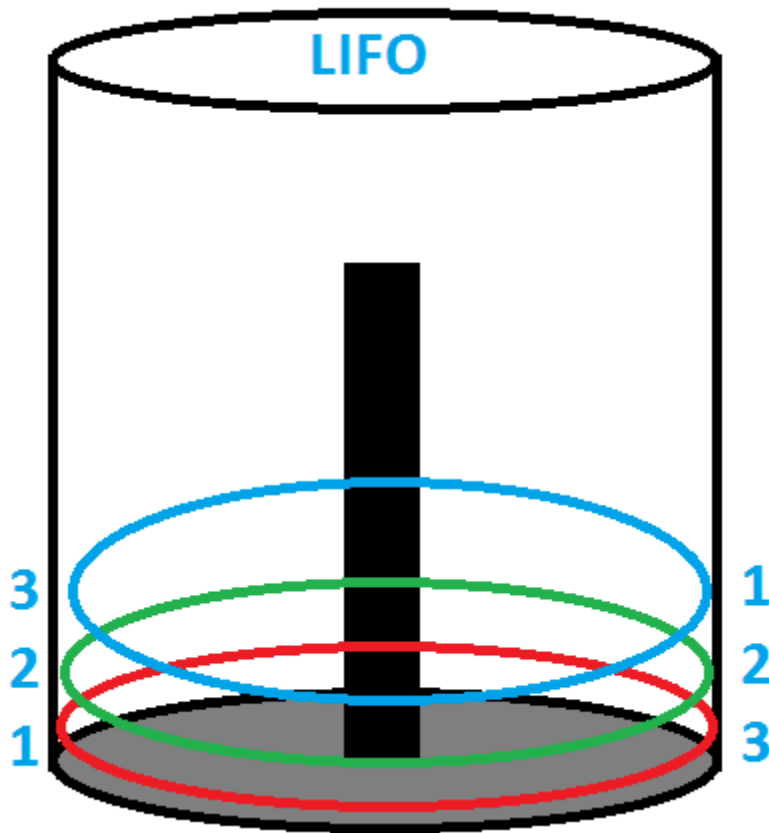
ram stack - 64kb
65536 bytes

$$64 * 1024 = 65536$$



LIFO-LAST IN FIRST OUT
FILO-FIRST IN LAST OUT





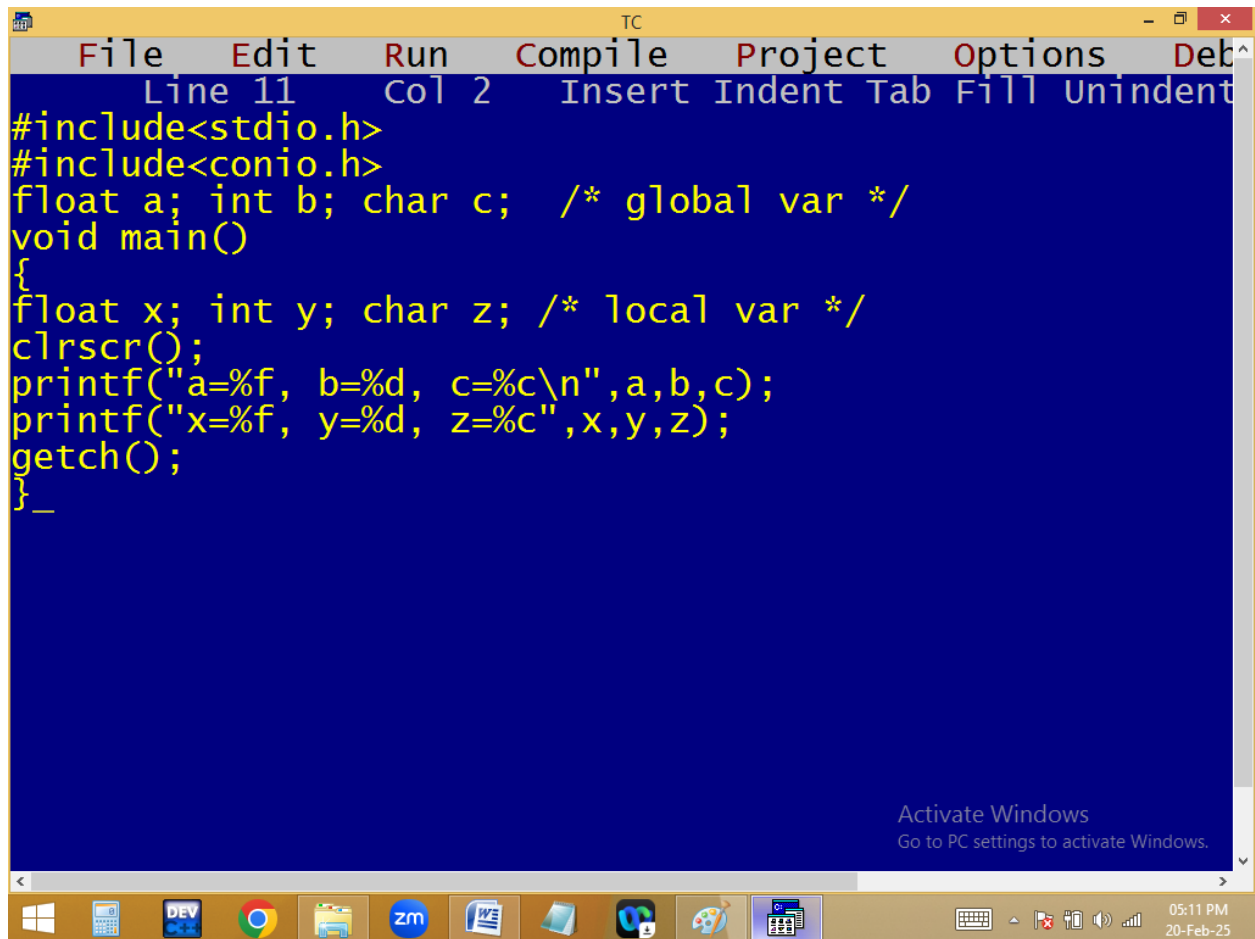
We are having 2 types of variables.

1. Local variables
2. Global variables

	Local variables	Global variables
Declaration	Within function	Top of program
Storage area	Stack area	Data area
Initial values	Garbage	Int=0, char=blank, float=0.000000

Scope-where we can access	Within fun only	Anywhere
Life time-until what time they are active in memory	Until function execution	Until total program execution

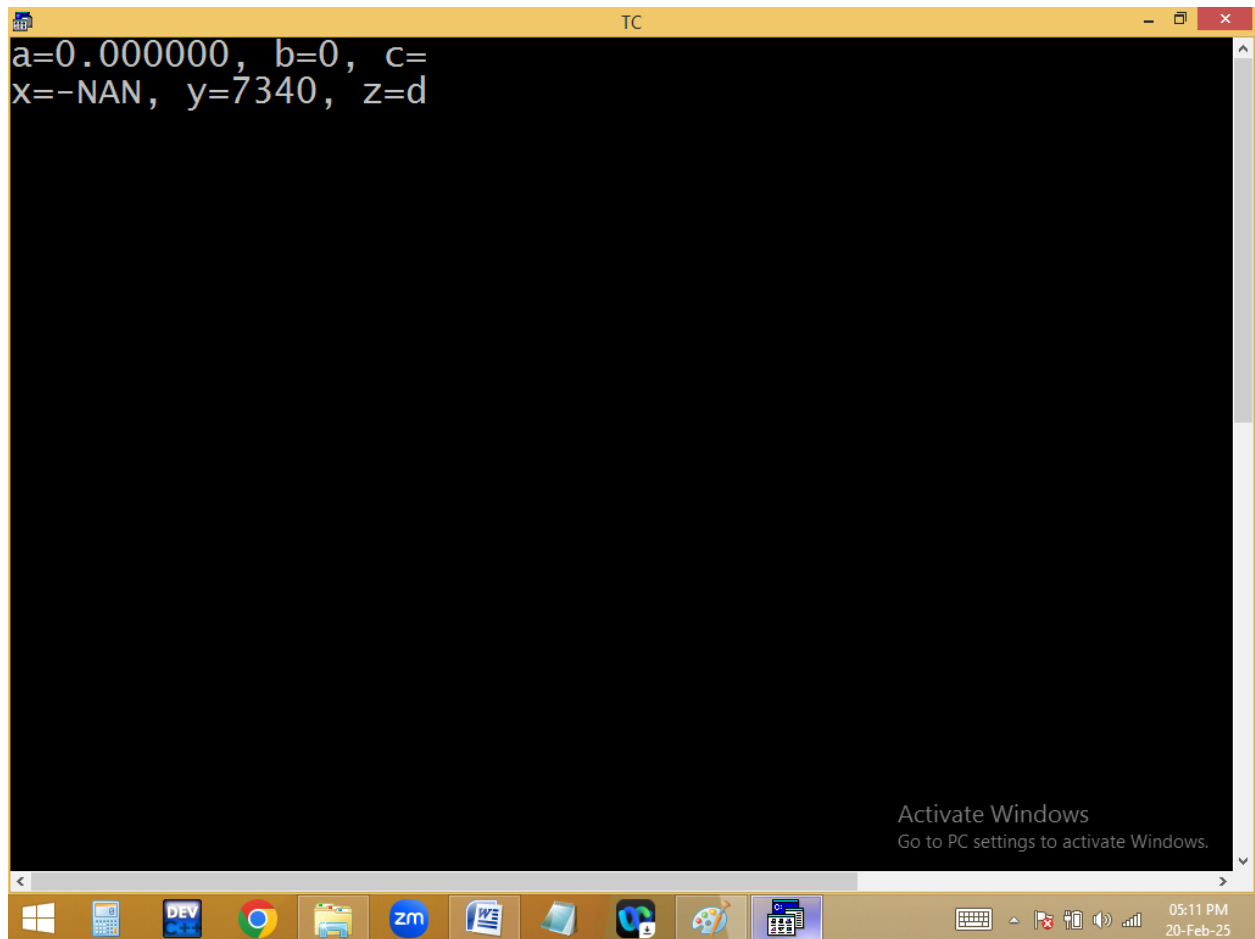
Finding initial values of local & global variables:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 11", "Col 2", and lists editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a blue background and contains the following C code:

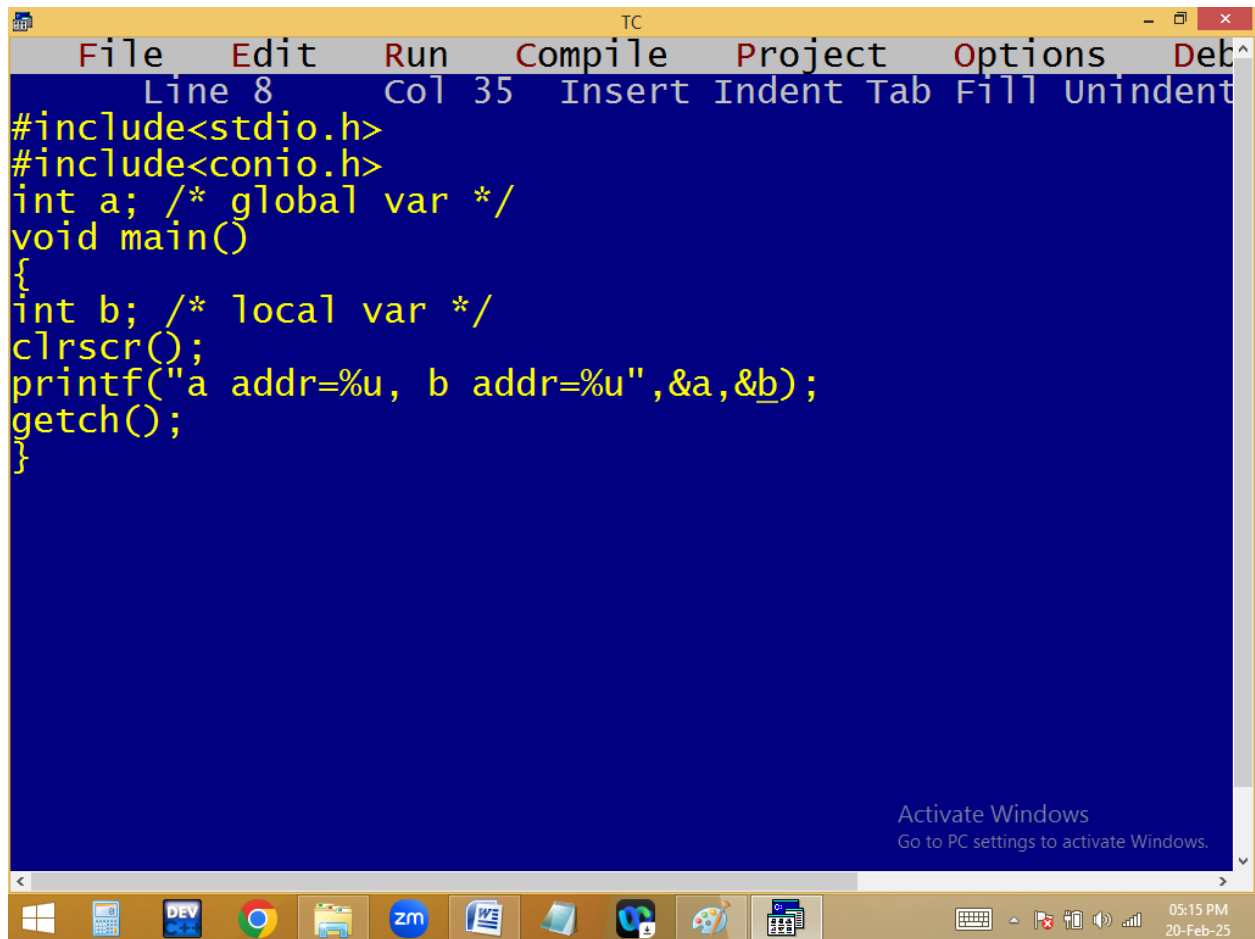
```
#include<stdio.h>
#include<conio.h>
float a; int b; char c; /* global var */
void main()
{
float x; int y; char z; /* local var */
clrscr();
printf("a=%f, b=%d, c=%c\n",a,b,c);
printf("x=%f, y=%d, z=%c",x,y,z);
getch();
}_
```

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows." The Windows taskbar is at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, a folder, OneDrive, Paint, and a calculator. The system tray on the right shows the date and time: "05:11 PM 20-Feb-25".



The image shows a screenshot of a Turbo C++ (TC) window. The window title bar is yellow and contains the text "TC". The main area of the window is black and displays the following text in white:
a=0.000000, b=0, c=
x=-NAN, y=7340, z=d
In the bottom right corner of the black area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons including Windows, Calculator, DEV, Chrome, File Explorer, Zoom, Word, and others. The system tray on the right shows the time as 05:11 PM and the date as 20-Feb-25.

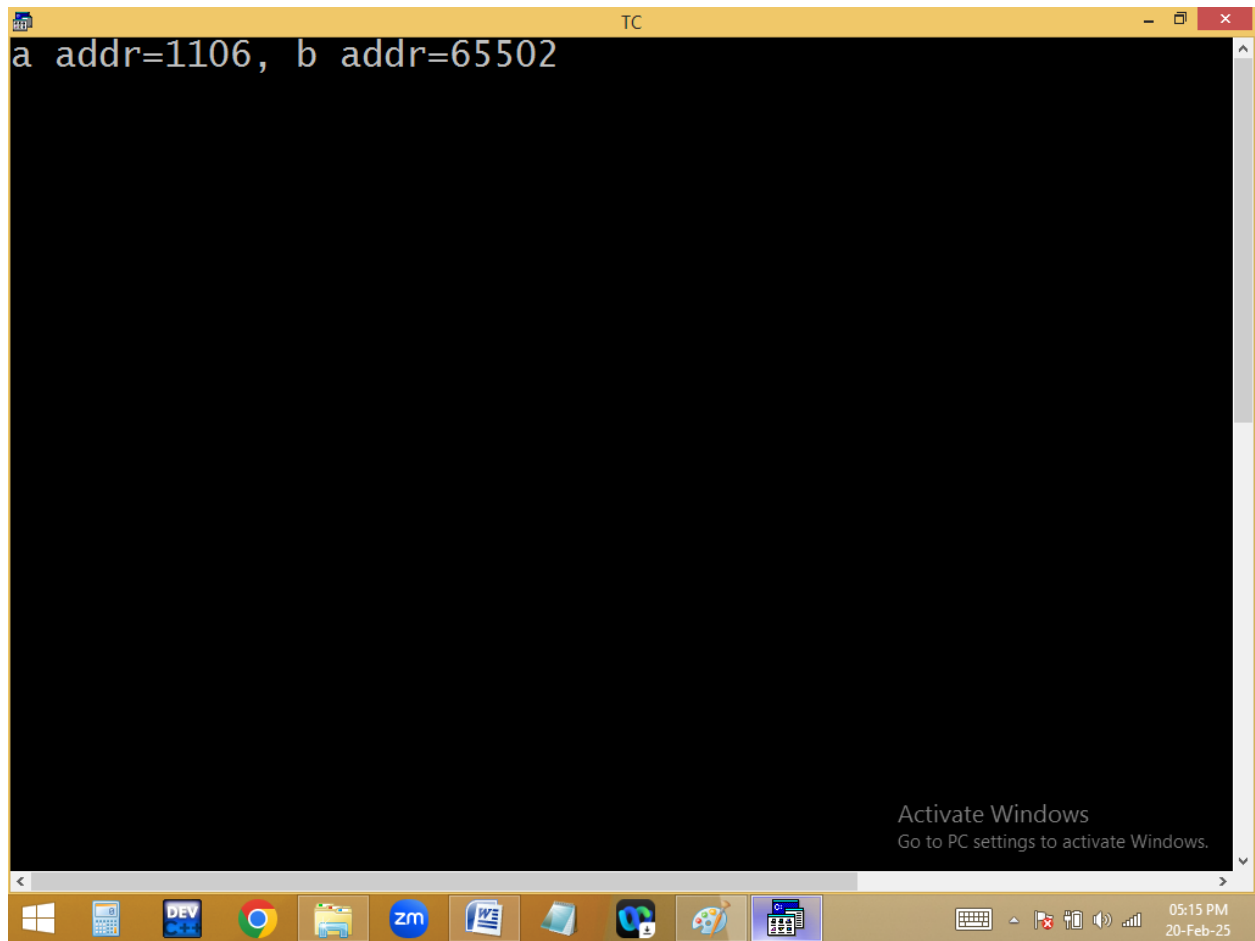
Finding storage area of local and global variables:



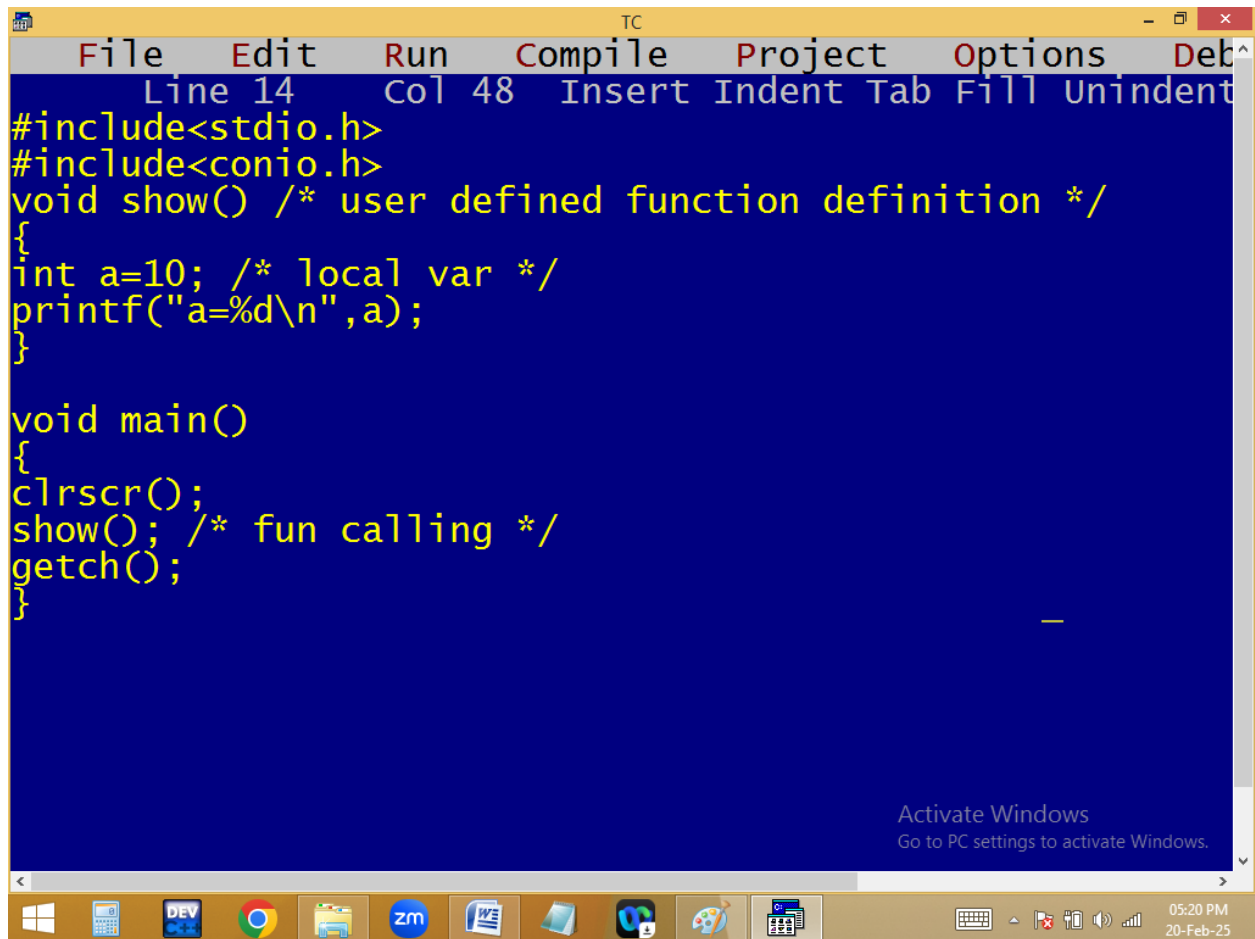
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 35", and a list of editing modes: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
int a; /* global var */
void main()
{
    int b; /* local var */
    clrscr();
    printf("a addr=%u, b addr=%u",&a,&b);
    getch();
}
```

In the bottom right corner of the IDE window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons including Windows, Calculator, DEV, Chrome, File Explorer, Zoom, Word, and others. The system tray on the right shows the time as 05:15 PM and the date as 20-Feb-25.



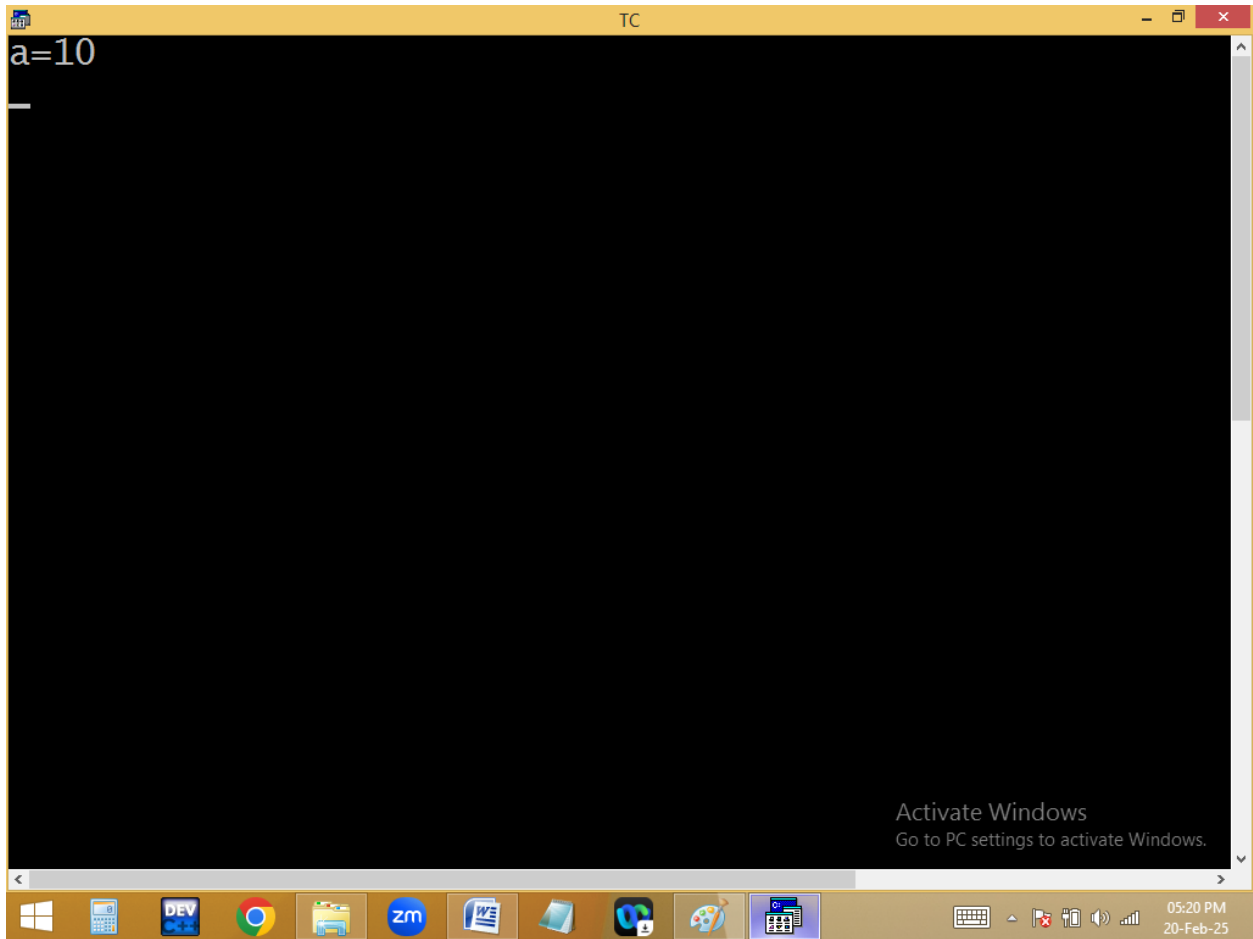
Finding scope of local variable:

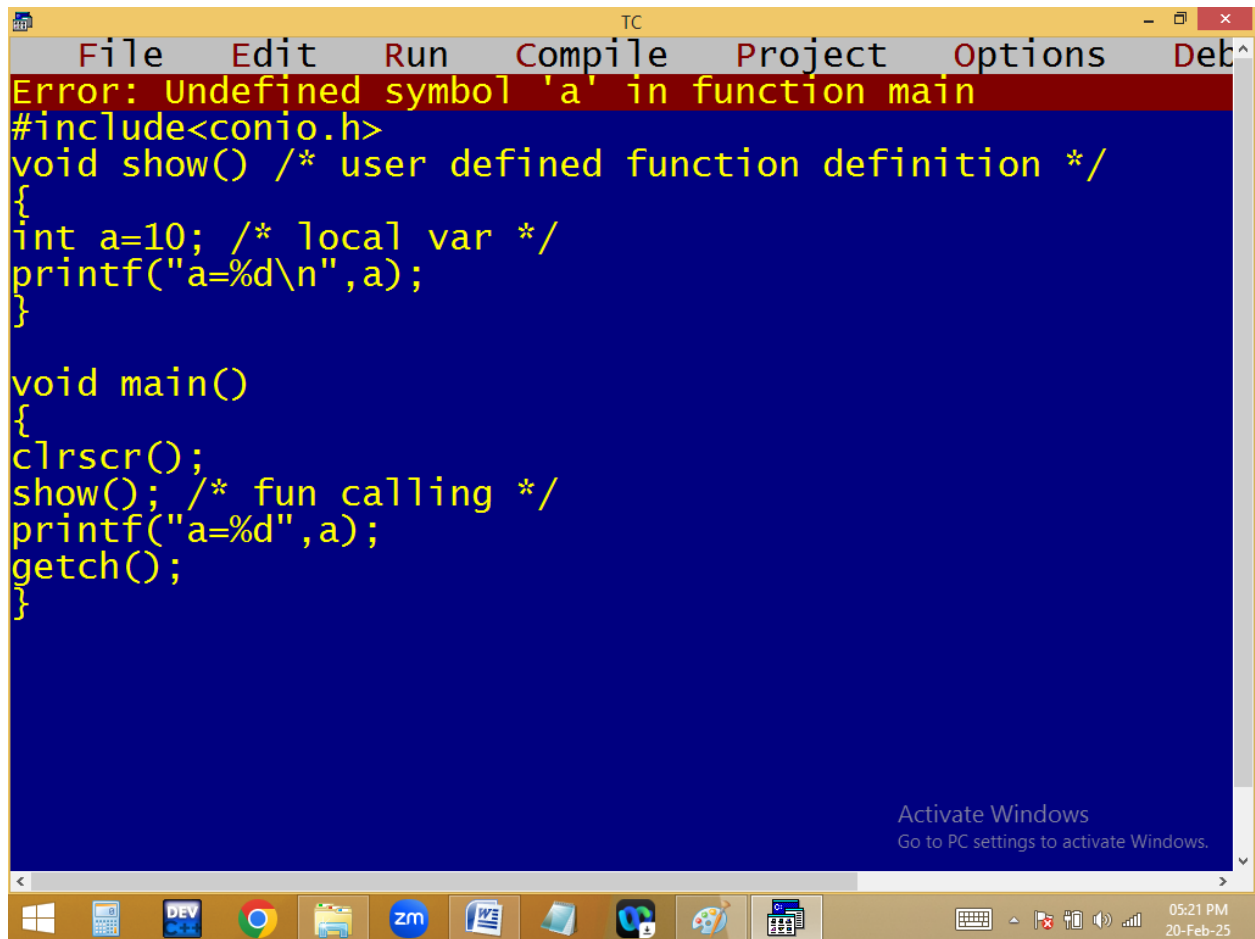


```
TC
File Edit Run Compile Project Options Deb
Line 14 Col 48 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void show() /* user defined function definition */
{
int a=10; /* local var */
printf("a=%d\n",a);
}

void main()
{
clrscr();
show(); /* fun calling */
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.





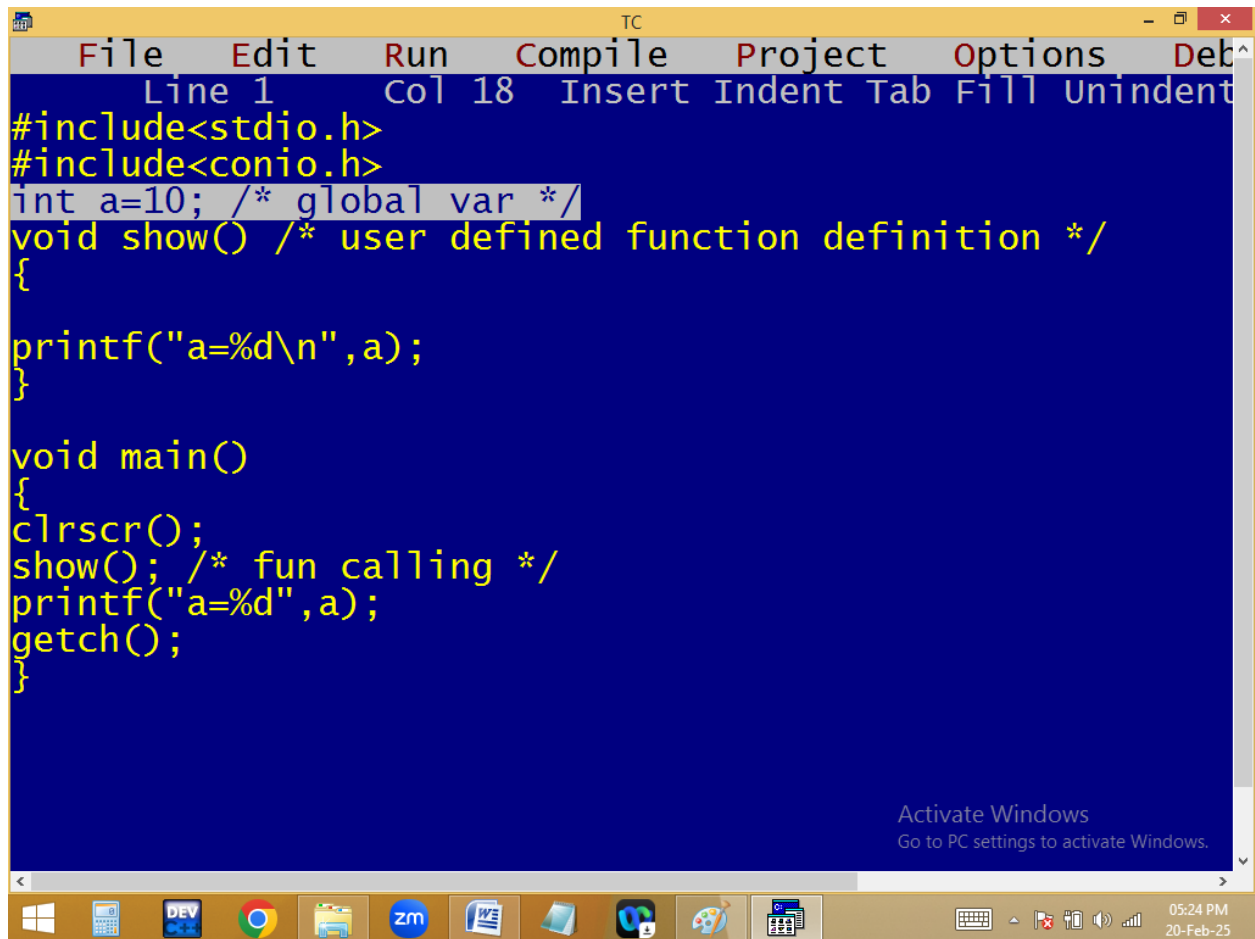
The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. A red error message banner at the top reads: "Error: Undefined symbol 'a' in function main". The code editor contains the following C code:

```
#include<conio.h>
void show() /* user defined function definition */
{
    int a=10; /* local var */
    printf("a=%d\n",a);
}

void main()
{
    clrscr();
    show(); /* fun calling */
    printf("a=%d",a);
    getch();
}
```

The Windows taskbar at the bottom shows various application icons including Windows Explorer, Dev C++, Google Chrome, and Zoom. The system clock in the bottom right corner displays "05:21 PM" and "20-Feb-25".

Finding global variable scope:

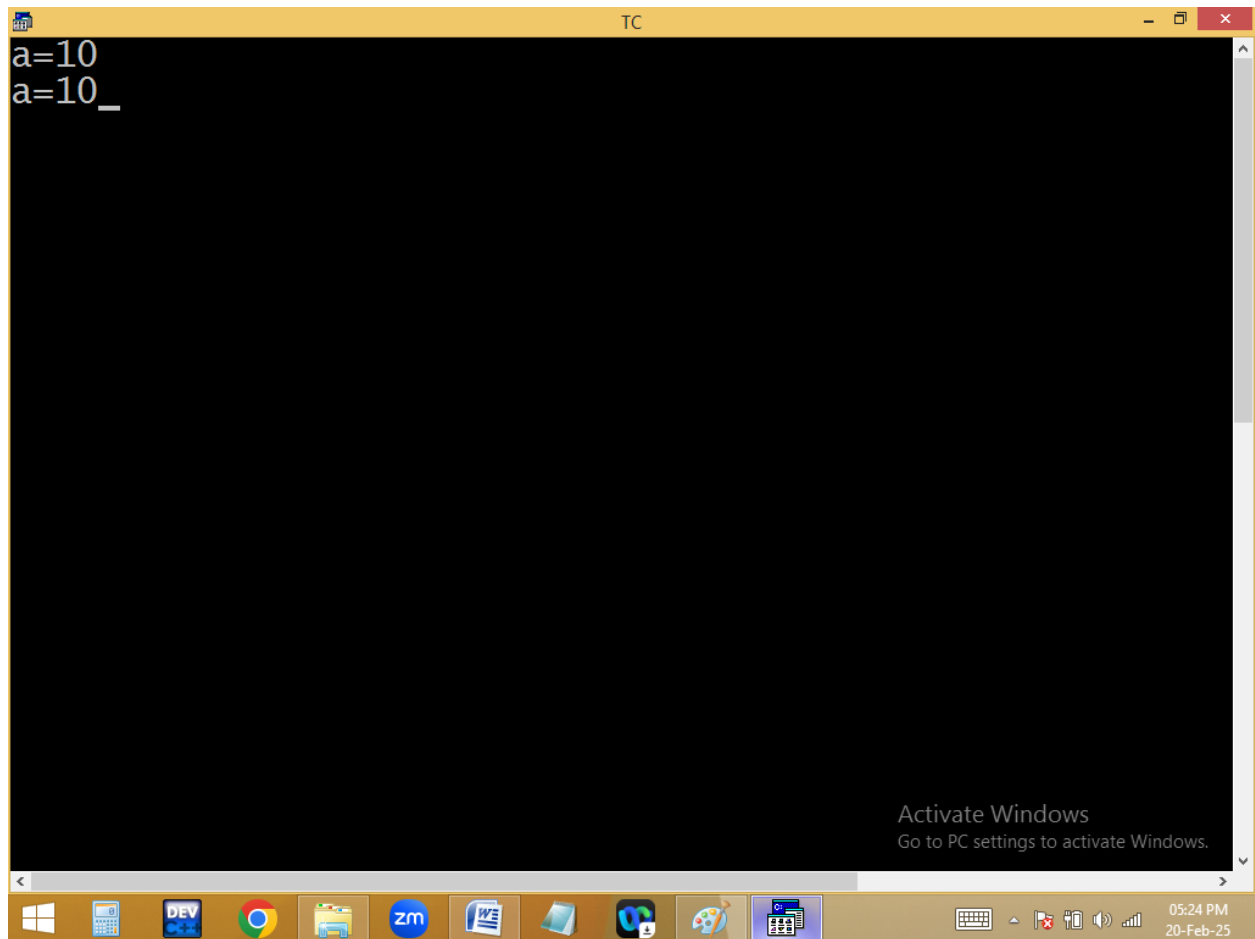


```
TC
File Edit Run Compile Project Options Deb^
Line 1 Col 18 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
int a=10; /* global var */
void show() /* user defined function definition */
{
printf("a=%d\n",a);
}

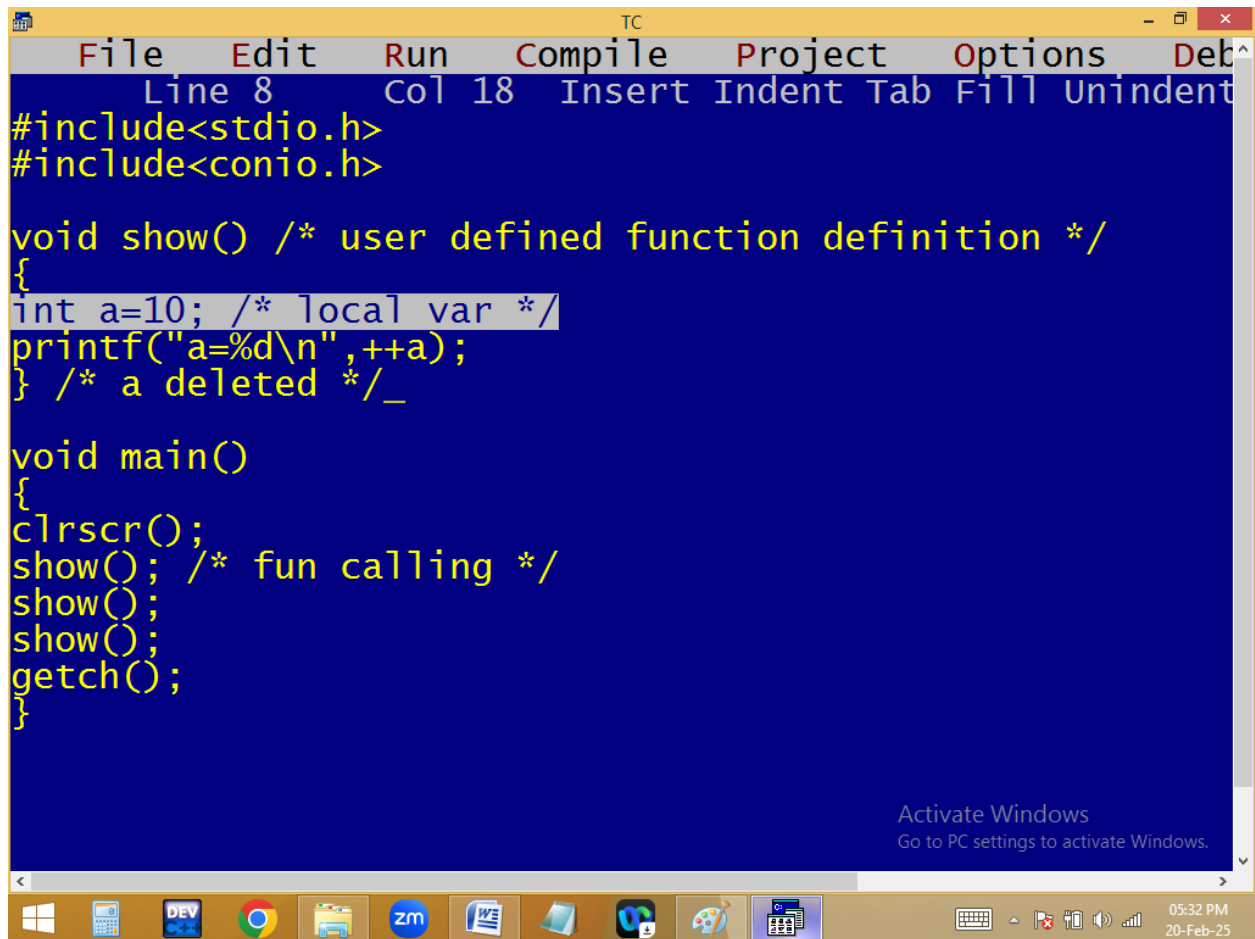
void main()
{
clrscr();
show(); /* fun calling */
printf("a=%d",a);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

05:24 PM
20-Feb-25



Finding life time of local variable:



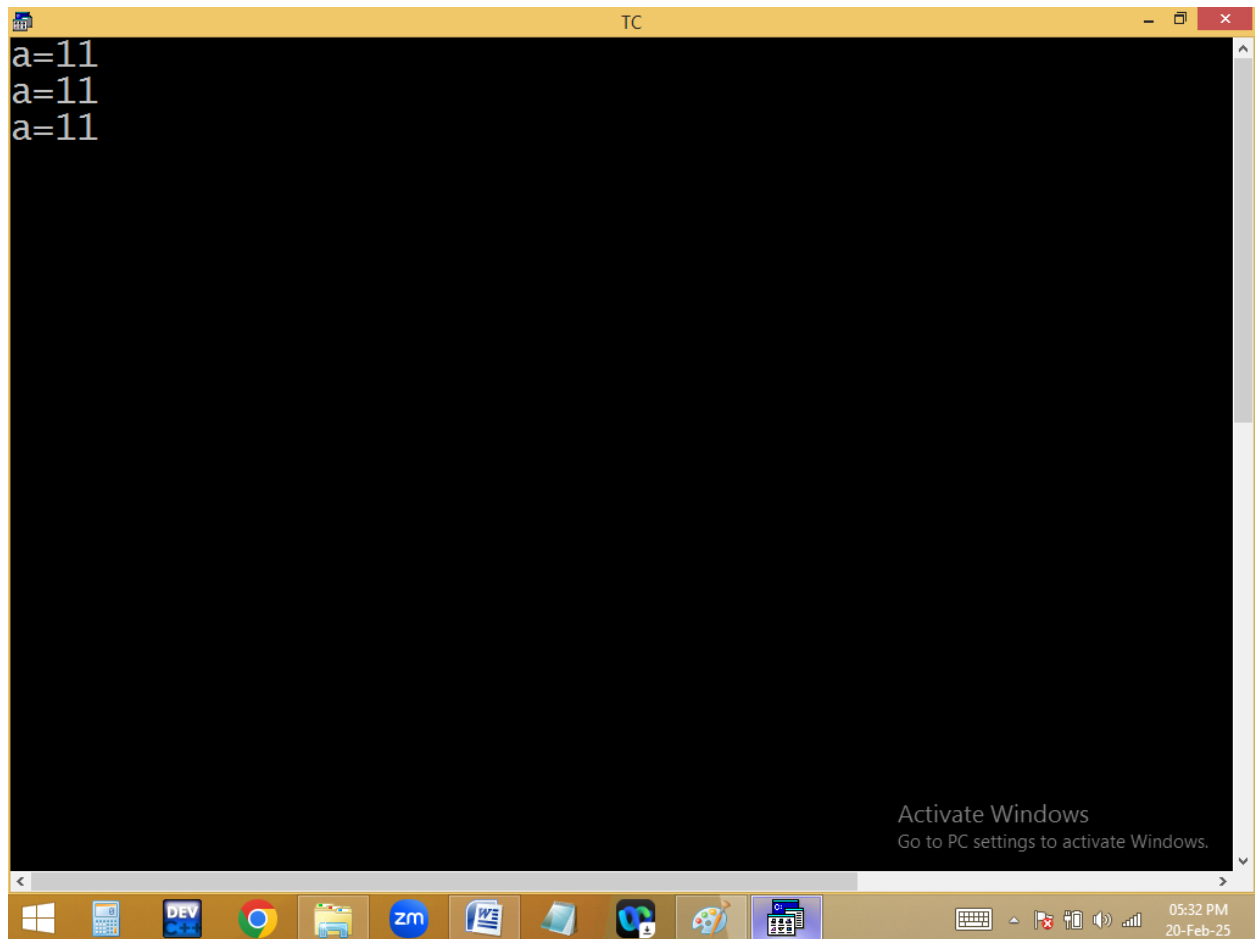
```
TC
File Edit Run Compile Project Options Deb^
Line 8 Col 18 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>

void show() /* user defined function definition */
{
int a=10; /* local var */
printf("a=%d\n",++a);
} /* a deleted */_

void main()
{
clrscr();
show(); /* fun calling */
show();
show();
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

05:32 PM
20-Feb-25

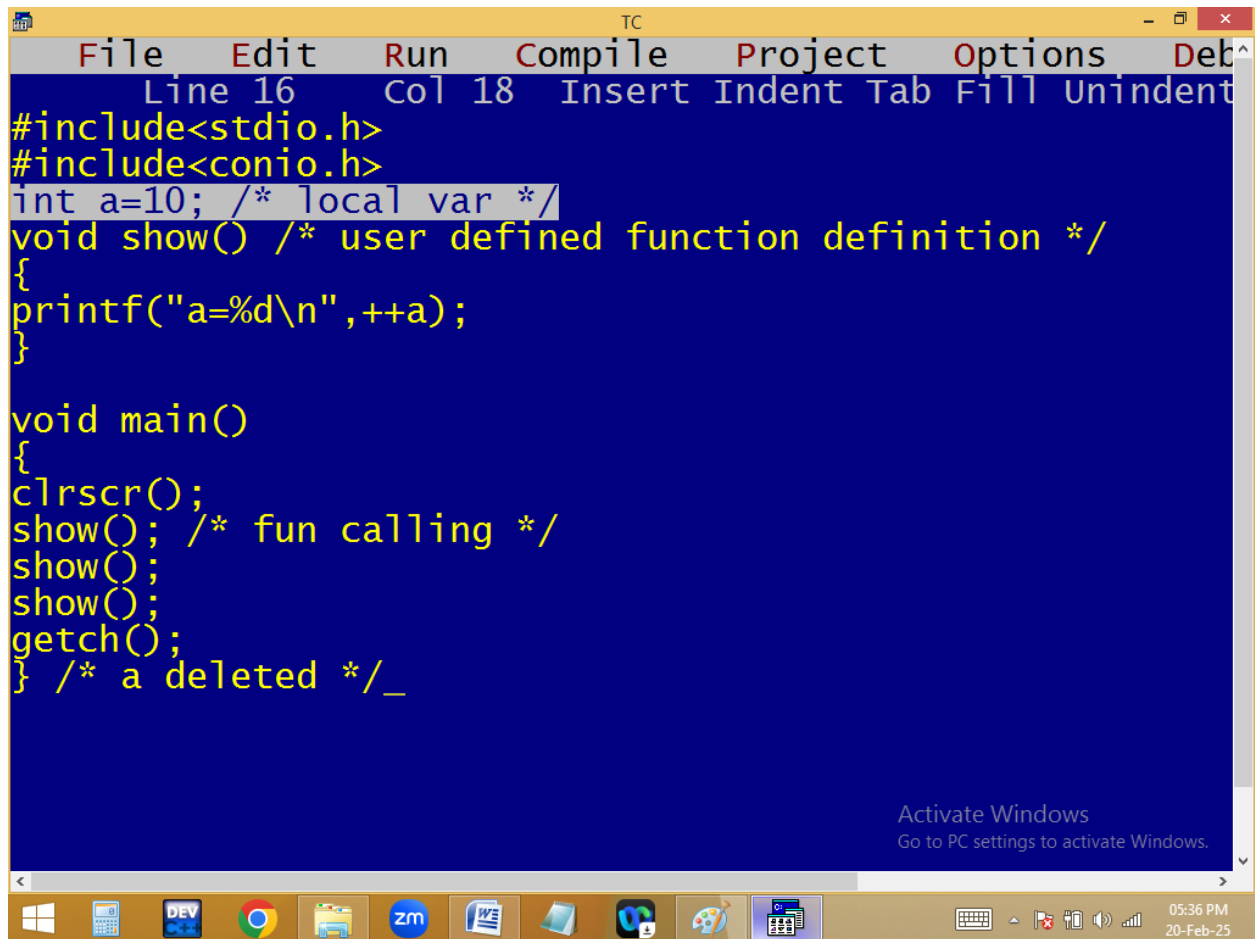


```
a=11
a=11
a=11
```

Activate Windows
Go to PC settings to activate Windows.

05:32 PM
20-Feb-25

Finding life time of global variable:

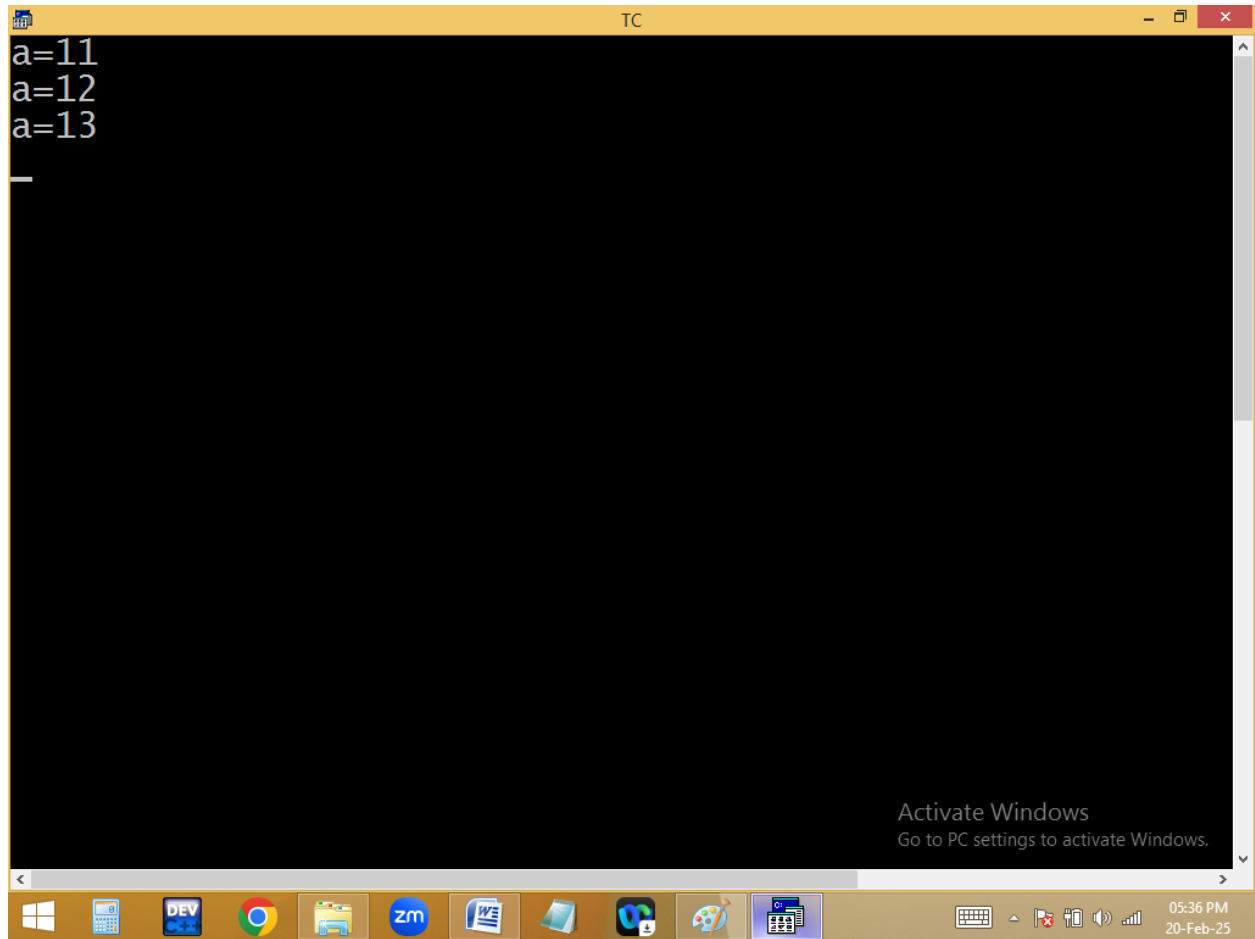


```
TC
File Edit Run Compile Project Options Deb^
Line 16 Col 18 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
int a=10; /* local var */
void show() /* user defined function definition */
{
printf("a=%d\n",++a);
}

void main()
{
clrscr();
show(); /* fun calling */
show();
show();
getch();
} /* a deleted */_

Activate Windows
Go to PC settings to activate Windows.

05:36 PM
20-Feb-25
```



Turbo C++ IDE

File Edit Search Run Compile Debug Project C

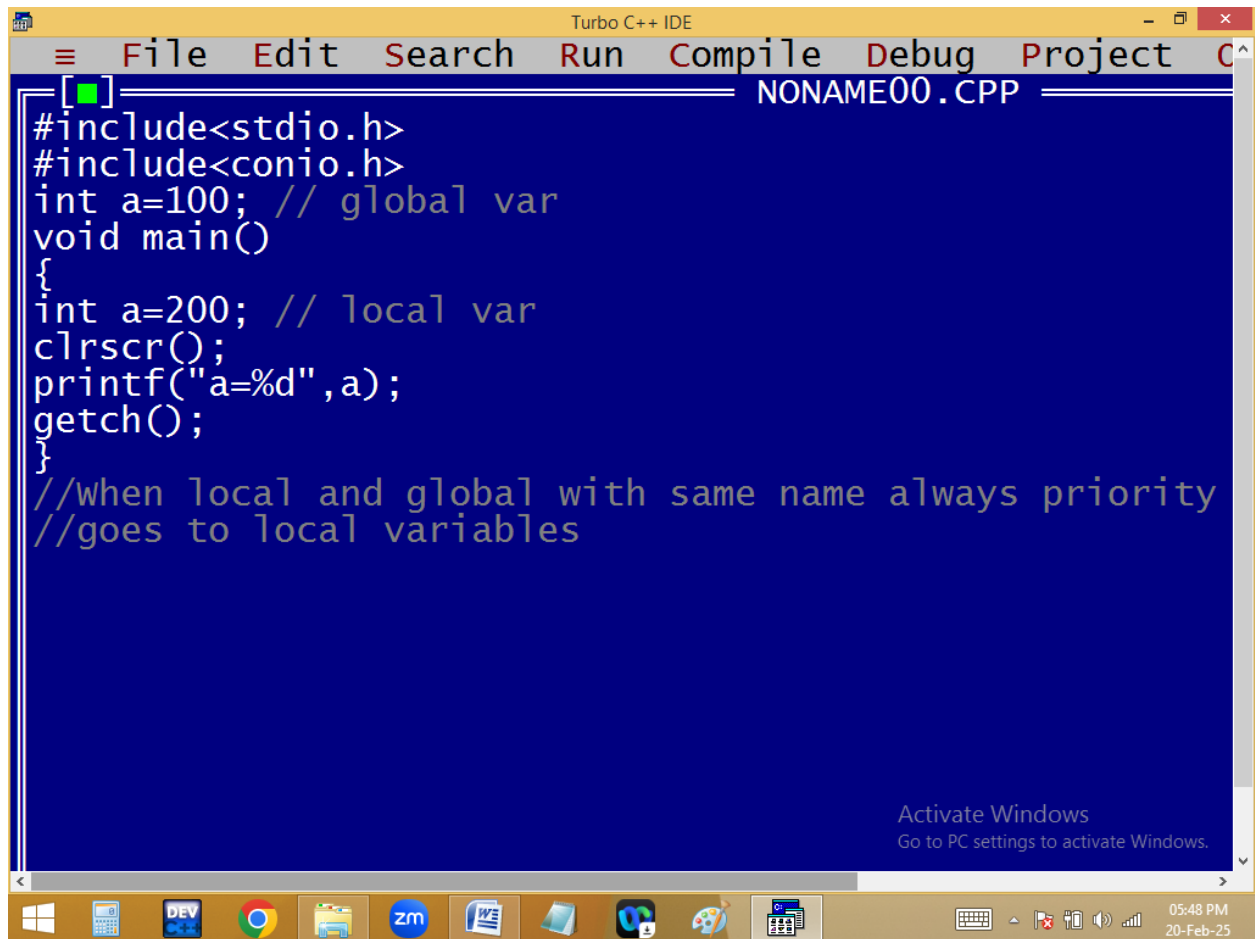
NONAME00.CPP

```
[■]  
#include<stdio.h>  
#include<conio.h>  
int a=100; // global var  
void main()  
{  
  int a=200; // local var  
  int a=300;  
  clrscr();  
  printf("a=%d",a);  
  getch();  
}
```

Activate Windows
Go to PC settings to activate Windows.

Error NONAME00.CPP 7: Multiple declaration for 'a'

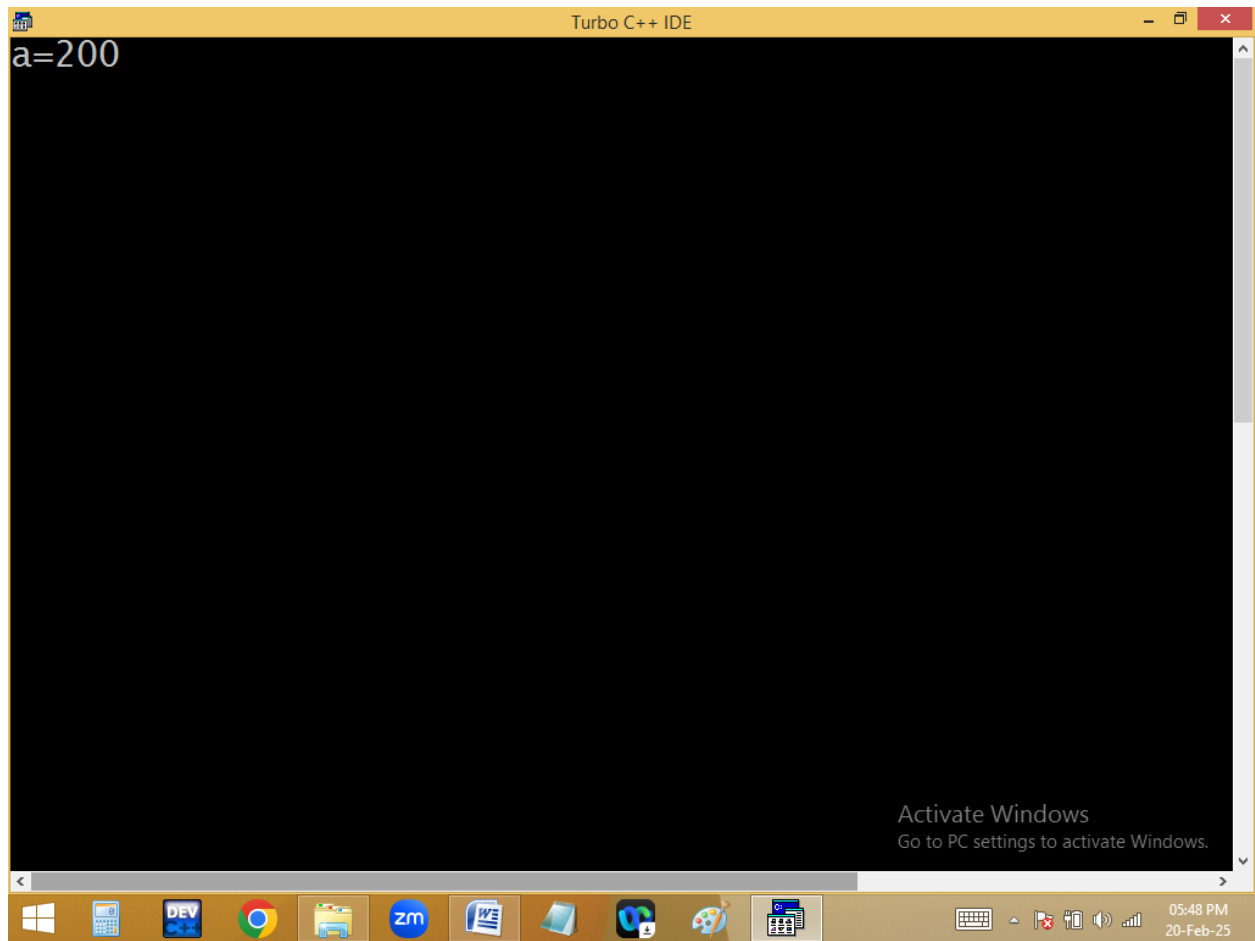
05:43 PM
20-Feb-25

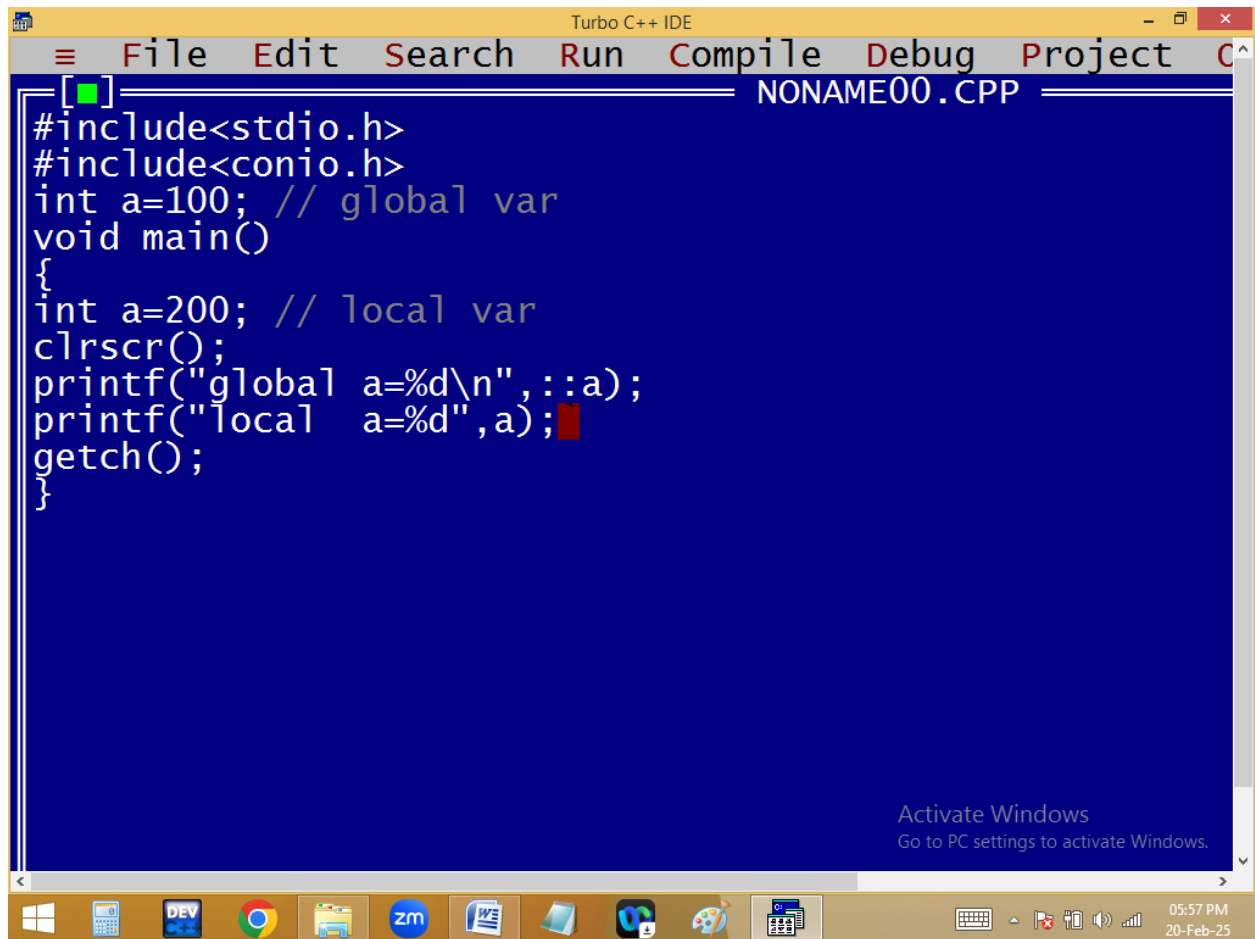


The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", and "Project". The file name "NONAME00.CPP" is displayed in the top right corner of the editor window. The code is written in C++ and demonstrates variable scope. It includes `<stdio.h>` and `<conio.h>`. A global variable `a` is declared and initialized to 100. The `main` function contains a local variable `a` initialized to 200, followed by `clrscr()`, `printf("a=%d",a);`, and `getch()`. Comments explain that local variables take priority over global variables with the same name. The Windows taskbar at the bottom shows various application icons and the system clock indicating 05:48 PM on 20-Feb-25.

```
[■] NONAME00.CPP
#include<stdio.h>
#include<conio.h>
int a=100; // global var
void main()
{
    int a=200; // local var
    clrscr();
    printf("a=%d",a);
    getch();
}
//when local and global with same name always priority
//goes to local variables
```

Activate Windows
Go to PC settings to activate Windows.





The image shows a screenshot of the Turbo C++ IDE. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", and "Project". The file name "NONAME00.CPP" is displayed in the top right corner of the editor window. The code in the editor is as follows:

```
[■]  
#include<stdio.h>  
#include<conio.h>  
int a=100; // global var  
void main()  
{  
  int a=200; // local var  
  clrscr();  
  printf("global a=%d\n",::a);  
  printf("local a=%d",a);  
  getch();  
}
```

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows." The Windows taskbar at the bottom shows the Start button, task view, and several application icons including DEV, Chrome, File Explorer, Zoom, Word, and others. The system tray on the right shows the date and time as "05:57 PM 20-Feb-25".

